

Implicit Methods

13

Formula for w_{i+1} depends on $f(\underline{w_{i+1}}, t_{i+1})$

↳ but we don't know this
so it leads to an algebraic equation

Adams - Moulton Methods

Two-stage (implicit) AM method

$$w_{i+1} = w_i + \frac{1}{12} h (5f(w_{i+1}, t_{i+1}) + 8f(w_i, t_i) - f(w_{i-1}, t_{i-1}))$$

$$\text{Local error } E = -\frac{1}{24} h^4 y^{(4)}(z) \text{ for some } z \in (t_{i-1}, t_{i+1})$$

Global error $O(h^3) \rightarrow$ third-order

Three-stage AM method

$$w_{i+1} = w_i + \frac{1}{24} h \left(9f(w_{i+1}, t_{i+1}) + 19f(w_i, t_i) - 5f(w_{i-1}, t_{i-1}) + f(w_{i-2}, t_{i-2}) \right)$$

Global error $O(h^4)$
↳ 4th order

One stage AM (backward Euler)

$$w_{i+1} = w_i + h f(w_{i+1}, t_{i+1})$$

Explicit vs. Implicit methods

* Implicit multistep methods have higher accuracy than explicit methods of the same order

⇒ example ^{Adams-Bashforth} A-B three stage $E = \frac{3}{8} y^{(4)}(z) h^4$ for some $z \in (t_{i-2}, t_{i+1})$
AM two stage $E = \frac{1}{24} y^{(4)}(z) h^4$ //

* Question: can we use an implicit method but reduce # of steps? // increase step-size

* Implicit methods are also better behaving in terms of stability. ↳ next topic

Predictor-corrector approach

Alternative to solving an algebraic equation

Use an estimate $\tilde{w}_{i+1}$ on the RHS of AM formula

$$\Rightarrow w_{i+1} = w_i + \frac{1}{12} h (5f(\tilde{w}_{i+1}, t_{i+1}) - \dots)$$

predictor
step

$$\tilde{w}_{i+1} = w_i + \frac{1}{2} h (3f(t_i, w_i) - \dots)$$

Use an explicit Adams-Bashforth as predictor
w/ the same number of stages (lower order)

* Using n -stage AB (order n) w/ an n -stage
AM (order $n+1$) has order $n+1$.

Fully implicit: solve Algebraic eqⁿ exactly // or numerically

If numerical $\rightarrow$ secant or Newton's

Stability Analysis

zero-stability $h \rightarrow 0$
absolute stability h fixed

15

To study stability, we analyze a simple ODE

$$\dot{y} = \lambda y \quad (\text{Dahlquist test condition})$$

$\lambda \in \mathbb{R}$ & $\lambda \leq 0$ typically

$$y(t) = y_0 e^{\lambda t} \rightarrow \text{if } \lambda < 0 \text{ sol}^n \text{ decays to zero}$$

Euler's method

$$w_{i+1} = w_i + h \underbrace{f(w_i, t_i)}_{\lambda w_i} = (1 + h\lambda) w_i$$

For $-1 < h\lambda < 0$ monotonic decrease

$-2 < h\lambda < -1$ oscillates & decreases

$-\infty < h\lambda < -2$ oscillates & grows $\rightarrow$

* Linear stability analysis

Method is unconditionally stable if $w_i \rightarrow 0$ as $i \rightarrow \infty$
for all $h\lambda < 0$

eg. Ralston's method (2nd order RK)

$$w_{i+1} = w_i + \frac{1}{4} h (\lambda w_i + 3\lambda (w_i + \frac{2}{3} h \lambda w_i))$$

$$= w_i + \alpha w_i + \frac{1}{2} \alpha^2 w_i = (1 + \alpha + \alpha^2) w_i$$

stable for $-2 \leq \alpha < 0$

Adaptive Stepsize

(17)

Estimating accuracy/error

⇒ We don't know the actual solⁿ but we know that an order- n method is more accurate than an order $n-1$ method.

For order- n $y(t+h) \approx y(t) + K_n$

For order $n-1$ $y(t+h) \approx y(t) + K_{n-1}$

$$|y(t+h) - y(t) - K_n| \ll |y(t+h) - y(t) - K_{n-1}|$$

$$\approx |y(t) + K_n - y(t) - K_{n-1}| = |K_n - K_{n-1}|$$

Tolerance ϵ means

$$|K_n - K_{n-1}| \leq \epsilon h$$

↳ bogatski
Bogacki-Shampine

⇒ MATLAB's ode23

3rd order Runge-Kutta with an embedded 2nd order method

↳ order —

→ If error $|K_n - K_{n-1}| > \epsilon h$ reject h & take a stepsize qh

If $|K_n - K_{n-1}| \leq \epsilon h$ accept qh

Pick q as $q = \left(\frac{\epsilon h}{|K_n - K_{n-1}|} \right)^{1/(n-1)}$